

Sustainable Pavements: Durability & Cost-Effectiveness of Plastic-Modified Asphalt in Southeastern Nigeria

Southeastern Nigeria's roads face rapid deterioration due to heavy rainfall, flooding, and problematic clay-rich subgrades. Traditional asphalt pavements fail prematurely, resulting in escalating maintenance costs and significant travel disruptions for motorists and communities.

Our research explores an innovative solution: converting plastic waste into asphalt modifiers. This approach offers dual benefits – extending pavement life while simultaneously reducing environmental plastic pollution that plagues our communities.

Presented by Engineer Abubakar Arome Muhammad from NARICT, this study examines how plastic-modified asphalt performs in our unique tropical climate and challenging soil conditions.

by ABUBAKAR AROME MUHAMMAD



Research Objectives & Questions

1 Quantify Performance Improvements

Measure and compare rut depth and crack density between plastic-modified and standard asphalt over an 18-month period to determine durability improvements under local conditions.

2 Compare Life-Cycle Costs

Analyze comprehensive cost factors, including initial construction expenses and ongoing maintenance requirements, to determine economic viability.

3 Assess Stakeholder Perspectives

Evaluate support levels and concerns among engineers, contractors, and motorists regarding the adoption of plastic-modified asphalt technology in southeastern Nigeria.

4 Research Questions:

- RQ1: How do rut depths and crack frequencies differ between the two pavement sections?
- RQ2: What is the total life-cycle cost per kilometer for each pavement type?
- RQ3: What benefits and risks do engineers and road users associate with using plastic-modified asphalt?
- RQ4: How willing are local stakeholders to support pilot projects using plastic-modified mixes?

Research Methodology

Field Experiment Setup

Construction of two adjacent 500m road sections near Enugu Airport - one using standard asphalt mix and one incorporating 6% shredded PET plastic waste.

Stakeholder Survey

Five-item questionnaire using Likert scale plus open comments distributed to 80 engineers, contractors, and motorists to gather perceptions.

Data Collection

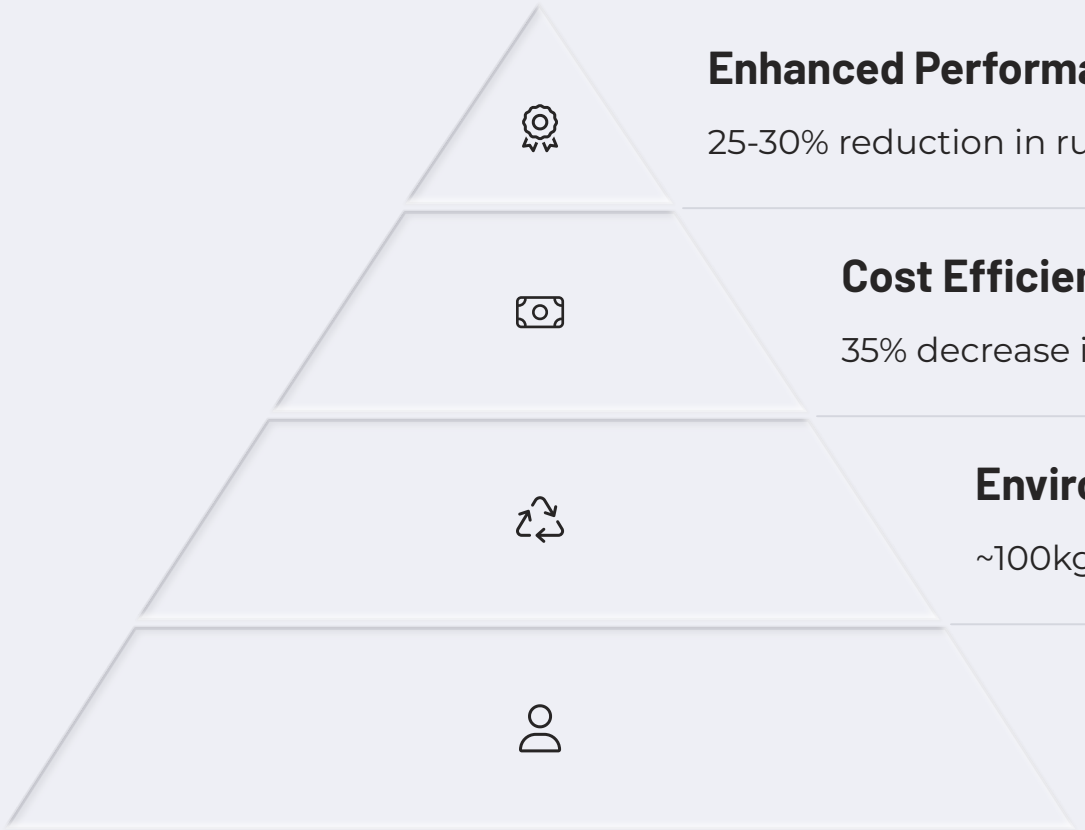
Measurements taken at 6, 12, and 18 months including rut depth (straightedge + ruler), crack density (visual inspection + grid sampling), and pothole count (number per km).

Analysis Methods

SPSS for descriptive statistics and t-tests comparing performance metrics and costs; NVivo for thematic coding of qualitative responses from surveys.

All research activities follow strict ethical guidelines including informed consent from participants, anonymity in reporting, and secure data storage protocols.

Expected Outcomes & Impact



Enhanced Performance

25-30% reduction in rut depths and crack formation

Cost Efficiency

35% decrease in three-year maintenance expenses

Environmental Benefits

~100kg PET waste diverted per km of road

Stakeholder Endorsement

Over 70% projected to support wider adoption

The findings will provide crucial evidence to inform state road agencies about the viability of plastic-modified asphalt. This research could potentially unlock TETFund or World Bank pilot grants to expand implementation across more Nigerian roads.

The dual benefit of addressing infrastructure durability while tackling plastic waste creates significant social and environmental value beyond the direct economic returns.

Conclusion & Recommendations

Pilot Expansion

Replicate trials across different road types and conditions throughout southeastern Nigeria to validate performance across diverse settings. Focus on high-traffic corridors connecting agricultural areas to markets.

Policy Engagement

Develop comprehensive guidelines and standards for plastic-asphalt blends specifically tailored to Nigerian conditions. Work with Federal Ministry of Works and Housing to incorporate findings into national specifications.

Capacity Building

Implement training programs for local contractors in proper blend preparation techniques and quality control measures to ensure consistent application and performance of plastic-modified asphalt.

Further Research

Investigate optimal plastic types and blend ratios for different Nigerian climate zones, focusing on maximizing durability while minimizing environmental impact through responsible sourcing.

Plastic-modified asphalt demonstrates clear durability and cost-effectiveness advantages in our tropical Nigerian context, offering a sustainable solution to both pavement distress and plastic pollution challenges.